



Physiological resolution of periodic breath holding during heavy-intensity Fartlek exercise

David J. Lim¹ · Jae J. Kim¹ · Greg D. Marsh¹ · Glen R. Belfry¹

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Abstract

Purpose The purpose was to compare the singular and combined effects of 5 s breath holds (BH) and 5 s sprints, every 30 s, during continuous high-intensity exercise, on ventilation ($\dot{V}_E$), oxygen uptake ($\dot{V}O_2$) and associated kinetics (τ), carbon dioxide production ($\dot{V}CO_2$), and arterialized-capillary lactate concentration ($[La^-]$).

Methods Ten men (24 ± 3 years) performed 4–6 min ergometer protocols that included a step-transition from 20 W to a power output of 50% of the difference between lactate threshold and $\dot{V}O_2$ peak ($\Delta 50\%$) including: (1) a continuous protocol (CONT) with free breathing, (2) an intermittent BH protocol (CONT-BH); repeated cycles of 5 s BH: 25 s free breathing, (3) a Fartlek protocol (Fartlek); repeated 5 s at peak aerobic power output: 25 s at $\Delta 50\%$; (4) combining the 5 s Fartlek and CONT-BH protocol (Fartlek-BH). Breath-by-breath gas exchange, measured by mass spectrometry and turbine, was recorded.

Results $\dot{V}_E$ ($L \min^{-1}$) was greater ($p < 0.05$) than CONT (90 ± 7) in all conditions CONT-BH (98 ± 16), Fartlek (105 ± 10), and Fartlek-BH (101 ± 19). $\dot{V}O_2$ and $\dot{V}CO_2$ ($L \min^{-1}$) were unchanged in CONT-BH (2.73 ± 0.14 and 3.16 ± 0.38) and greater in Fartlek (2.85 ± 0.12 and 3.43 ± 0.16), compared to CONT (2.71 ± 0.12 and 3.12 ± 0.13). Whereas, $\dot{V}CO_2$ during Fartlek-BH was higher (3.28 ± 0.35) and $\dot{V}O_2$ was unchanged (2.73 ± 0.14). Fartlek-BH resulted in slower $\dot{V}O_2$ kinetics (62.2 ± 19 s) and greater blood lactate concentrations (11.5 ± 2.7 mM), compared to CONT (48.8 ± 12 s; 9.0 ± 2.3 mM, respectively).

Conclusions It was demonstrated that the CONT-BH resulted in increased ventilation that sustained $\dot{V}O_2$. However, $\dot{V}O_2$ was restricted when an additional work was combined with the BH condition.

Keywords O_2 uptake · Breath holding · Muscle deoxygenation · Intermittent exercise · Swimming · Backstroke

Abbreviations

BH	Breath hold
CO_2	Carbon dioxide
CONT	Continuous exercise
CONT-BH	Continuous breath hold
Fartlek-BH	Fartlek breath hold
H^+	Hydrogen ion
$[Hb_{tot}]$	Total hemoglobin concentration
$[HHb]$	Deoxygenated hemoglobin concentration
$[HHb]/\dot{V}O_2$	Adjustment of normalized $[HHb]$ -to- $\dot{V}O_2$ ratio
$[La^-]$	Arterialized-capillary lactate concentration

LT	Estimated lactate threshold
min	Minute
ms	Millisecond
N_2	Nitrogen
NIRS	Near-infrared spectroscopy
O_2	Oxygen
PCr	Phosphocreatine
PCO_2	Partial pressure of carbon dioxide
$P_{ET}CO_2$	End-tidal partial pressure of carbon dioxide
$P_{ET}O_2$	End-tidal partial pressure of oxygen
PO	Power output
PO_2	Partial pressure of oxygen
RPM	Revolutions per minute
s	Second
$\pm$	Standard deviation
$S_{at}O_2$	Tissue hemoglobin saturation
τ	Time constant
$\dot{V}CO_2$	Carbon dioxide production
$\dot{V}_E$	Minute ventilation
$\dot{V}O_2$	Oxygen uptake

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✉ Glen R. Belfry
 gbelfry@uwo.ca

¹ School of Kinesiology, Faculty of Health Sciences, Canadian Center for Activity and Aging, University of Western Ontario, 1151 Richmond Street, London, ON, Canada

$\dot{V}O_{2\text{peak}}$	Peak oxygen uptake
W	Watts
μ_A	Absorption coefficient
μ_S	Scattering coefficient

Introduction

Many sports such as cycling (Paterson et al. 1986), rowing (Siegmund et al. 1999), and swimming (Dicker et al. 1980) require brief intermittent periods of high power output in both training and competition. There are also instances where an athlete must produce higher power outputs while breathing irregularly or not breathing at all. For example, breath holds during competitive backstroke swimming are required as the swimmers push off the wall and kick underwater after a turn. In a 50 m pool, these underwater, kicking only phases, may be performed at various intensities for a maximum of 15 m that endure for ~5 s (Veiga et al. 2015). After surfacing, these swimmers are able to breathe freely as they complete the remaining distance in ~25 s while using the upper and lower limbs to swim the full stroke (Skorski et al. 2013). Little is known about the cardiorespiratory consequences of this pattern of intermittent breath holding during high-intensity exercise.

The previous research has shown that long-duration breath holding during exercise induces hypoxia and hypercapnia. As a result, decreases in alveolar and arterial partial pressures of O_2 (PO_2 ; hypoxemia) and increases in CO_2 (PCO_2 ; hypercapnia) have been observed (Ward and Nguyen 1991; Kume et al. 2013). This caused breath hold-induced decreases in O_2 delivery and pulmonary oxygen uptake ($\dot{V}O_2$) (Andersson et al. 2004), as well as increases in muscle deoxygenation ([HHb]) (Woorons et al. 2011; Kume et al. 2013) and arterialized-capillary lactate concentrations ($[La^-]$), compared to free-breathing continuous exercise (Hoffmann et al. 2005; Wein et al. 2007; Whipp and Davis 1979). It has also been shown that at the onset of square-wave constant-load exercise under hypoxic conditions, above and below the lactate threshold, the rate of adjustment of pulmonary oxygen uptake ($\dot{V}O_2$ kinetics) is slowed (Engelen et al. 1996; DeLorey et al. 2004; Hughson and Kowalchuk 1995; Springer et al. 1991).

It was postulated that regular insertion of the aforementioned shorter breath hold durations (5 s) for every 30 s would also elicit changes in $P_{ET}O_2$, $P_{ET}CO_2$, [HHb] and $[La^-]$. However, a compensatory increase in $\dot{V}_E$ during the 25 s free-breathing periods (Whipp 1977) would mitigate the previously observed breath hold-induced hypoxemia, hypercapnia, increased $[La^-]$, and slowed $\dot{V}O_2$ kinetics.

Intermittent exercise studies have shown that reducing the mean power output by including regular 5 s recovery periods within heavy-intensity exercise results in decreased

mean $\dot{V}O_2$ and [HHb]. Temporal reductions in both variables during the 5 s recovery periods pursuant to the decreased O_2 demand were also observed (Belfry et al. 2012a). If the 5 s recovery periods were replaced by 5 s sprints, as may be performed within the aforementioned underwater kicking model, it is suggested that temporal changes in $\dot{V}O_2$ and [HHb] would also result.

Moreover, changes in mean power output during continuous exercise, across the intensity continuum from 90% of lactate threshold to 110% of $\dot{V}O_2$ peak, have resulted in similar time constants for the on-transient kinetics of $\dot{V}O_2$ (τ) and decreased $[La^-]$ (Ozyener et al. 2001). Tau being defined as the time to reach 63% of the zenith of the mono-exponential $\dot{V}O_2$ response that is observed soon after the abrupt application of a sub-lactate threshold load (moderate intensity work). For power outputs above the lactate threshold (heavy-intensity work), such as were prescribed in the present study, this initial mono-exponential $\dot{V}O_2$ response is followed by a slowly developing second exponential (Ozyener et al. 2001). At these heavy-intensity work rates, τ is calculated upon reaching the zenith of the first exponential before the onset of the second (Ozyener et al. 2001). Consequently, it is suggested that increasing the mean power output during constant-load high-intensity exercise by inserting regular 5 s sprints (Fartlek) would leave $\dot{V}O_2$ kinetics (τ) unchanged. However, increases in $\dot{V}_E$, mean $\dot{V}O_2$, [HHb], and $[La^-]$ would be observed. Within the context of the backstroke swimming model, this perturbation would enable the physiological responses to different intensities of kicking to be studied.

If these 5 s sprints were combined with breath holds, it is proposed that the reduction of breathing opportunities would overwhelm any ventilatory and/or HHb attempt to maintain O_2 availability at the level of the muscle to support the increased O_2 demand, and $\dot{V}O_2$ kinetics (τ) would slow (Spencer et al. 2012). Moreover, the ventilatory buffering-related $\dot{V}CO_2$ would be tempered resulting in an increase in $[La^-]$ (Rossiter et al. 1999).

Without a swimming flume or underwater compatible gas mass spectrometry and muscle near-infrared spectroscopy collection systems available, coupled with the distinct and simultaneous physiological demands of the upper and lower limbs during backstroke swimming (Jones et al. 2014), this initial study, determining the physiological effects of 5 s breath holds, with and without regular 5 s sprints, will limit its focus to the cardiorespiratory responses of the lower limbs during cycle ergometry.

The purpose of this study was to compare and contrast $\dot{V}O_2$, $\dot{V}O_2$ kinetics, muscle deoxygenation, and blood lactate responses during continuous heavy-intensity cycling exercise with free-breathing, to three differing exercise conditions: (1) continuous heavy-intensity exercise incorporating repeated 5 s breath holds every 30 s, (2) repeated 30 s cycles comprised of 25 s of this same heavy-intensity exercise

followed by 5 s of higher power output with free-breathing throughout, and (3) combining the 5 s breath holds with 5 s of higher power output during heavy-intensity exercise.

It was hypothesised that, in comparison to continuous exercise: (1) the repeated 5 s breath hold condition would result in similar mean $\dot{V}O_2$ and $\dot{V}O_{2\text{peak}}$ kinetics; (2) the repeated 5 s sprints would result in increased mean $\dot{V}O_2$ and $\dot{V}_{E\text{peak}}$, as well as unchanged $\dot{V}O_2$ kinetics and $[La^-]$. However, (3) when the two perturbations were combined, the aforementioned physiological resolution would be overwhelmed, resulting in unchanged mean $\dot{V}O_2$ and $\dot{V}_{E\text{peak}}$, as well as increased $[La^-]$ and slower $\dot{V}O_2$ kinetics.

Methods

Participants

Ten adult males (age 24 ± 3 year; height 179 ± 9 cm; weight 80 ± 9 kg) volunteered and gave written consent to participate in this study. All procedures were approved by the Western University Research Ethics Board for Health Sciences Research Involving Human Participants and were in accordance with the 1964 Helsinki declaration and its later amendments or comparable ethical standards. All participants gave their informed consent and were healthy, recreationally active (active 1–3 times per week), and non-smokers. None of the participants were taking any medications that would affect the cardiorespiratory or hemodynamic responses to exercise.

Testing protocol

Participants were asked to maintain their usual levels of physical activity throughout their participation in the present study and to refrain from drinking caffeine 6 h prior to their tests. All tests were performed on an electronically braked cycle ergometer on 5 separate days. All tests were performed within a 48 ± 2 h period after each test. During each test, the participants were required to wear a nose clip to prevent the participant from breathing through their nose, and a rubber mouthpiece, similar to that of breathing through a snorkel or diving mask.

Testing Day 1

An incremental ramp test to fatigue on a cycling ergometer with a work rate increment of 25 W min^{-1} was performed with verbal encouragement to facilitate peak efforts. Participants were instructed to maintain a cadence of 70 revolutions per min (RPM). The test was terminated when the participant could no longer maintain 60 RPM. This test took approximately 15 min to complete and was used to determine the $\dot{V}O_{2\text{peak}}$ and the estimated lactate threshold (LT). These measures were then used to prescribe the work rate for the heavy-intensity and peak aerobic power outputs in subsequent tests for each individual. The LT was defined as the $\dot{V}O_2$ at which $\dot{V}CO_2$ (carbon dioxide production as measured in the pulmonary space) began to increase out of proportion to $\dot{V}O_2$ with a systematic rise in minute $\dot{V}E$ (V_E)-to- $\dot{V}O_2$ ratio and end-tidal PO_2 , whereas the V_E -to- $\dot{V}CO_2$ ratio and end-tidal PCO_2 were stable (Beaver et al. 1981). The LT was determined through visual inspection by two researchers familiar with this procedure. Data analysis of the LT began after accounting for the delay between $\dot{V}O_2$ and the change in work rate (the cardiodynamic phase) during the incremental ramp test to fatigue. These data are presented in Table 1.

Testing Day 2

Participants performed a ‘square-wave’ cycling exercise test. This constituted a 4 min warm-up (20 W) followed by an instantaneous power output increase to a heavy-intensity work rate for 6 min while free-breathing (CONT). The power output at this heavy-intensity work rate corresponded to the work rate at which the participant’s $\dot{V}O_2$ was at 50% of the difference between the $\dot{V}O_2$ at LT and $\dot{V}O_{2\text{peak}}$ ($\Delta 50\%$) during their incremental ramp test. Participants were instructed to maintain a pedal cadence of 70 RPM during all the tests.

Testing Day 3

Participants performed the identical ‘square-wave’ cycling exercise test as Testing Day 2. However, repeated cycles of 25 s of free breathing, followed by 5 s of breath holding (CONT-BH), were performed from the beginning of the warm-up until the end of the 6 min heavy-intensity work rate.

Table 1 Participant characteristics and performance variables from the ramp incremental test including $\dot{V}O_{2\text{peak}}$, estimated LT, peak PO, and PO at $\Delta 50$

	$\dot{V}O_{2\text{peak}}$ (L min^{-1})	$\dot{V}O_{2\text{peak}}$ ($\text{mL kg}^{-1} \text{ min}^{-1}$)	Estimated LT (L min^{-1})	PO at $\Delta 50$ (W)	Peak PO (W)
Mean	3.17	40	1.77	218	314
SD	0.52	6	0.22	30	49

$\dot{V}O_{2\text{peak}}$ peak oxygen uptake, *Estimated LT* estimated lactate threshold, *PO at $\Delta 50$* power output at 50% of the difference between the LT and $\dot{V}O_{2\text{peak}}$, *Peak PO* peak power output attained during ramp incremental test (PO used for the sprints)

To ensure proper execution, participants were given a 5 s verbal countdown leading into each breath hold. Participants were also instructed to regulate their breathing during this 5 s period to ensure that the inspiration initiating the breath hold was completed at time zero.

Testing Day 4

Participants performed the identical 3 min warm-up followed by the instantaneous increase to the heavy-intensity work rate on the cycle ergometer. However, the 6 min of heavy-intensity work was composed of repeated 30 s cycles, comprised of 25 s at the same heavy-intensity work rate ($\Delta 50\%$) as the previous tests, followed by 5 s at the peak power output (Fartlek) attained during the ramp incremental test. Non-regulated breathing was performed throughout this test.

Testing Day 5

Participants performed the identical cycle ergometer protocol as Testing Day 4. However, repeated cycles of 25 s of non-regulated breathing, followed by 5 s of breath holding, were performed from the beginning of the warm-up until the end of the 6 min Fartlek protocol (Fartlek-BH).

The order of performing the four conditions (CONT, CONT-BH, Fartlek, and Fartlek-BH) was systematically randomized via the iPhone application. The iPhone application is called “random”. A “sequence generator” was used. This application is available at <https://www.random.org/sequences/?min=1&max=4&col=1&format=html&rnd=new>

Measurements

Inspired and expired gases were measured breath-by-breath by utilizing a mass spectrometer (Innovision, AMIS 2000, Lindvedvej, Denmark). Gas collection for inspired and expired flow rates was also measured with a low-dead-space (90 mL) bidirectional turbine (Alpha Technologies, VMM 110) and pneumotach (Hans Rudolph, Model 4813) which were calibrated before each test using a syringe of known volume (3 L). Inspired and expired gases were sampled continuously at the mouth and analyzed for concentrations of O_2 , CO_2 , and N_2 by mass spectrometry after calibration with precision-analyzed gas mixtures. Changes in gas concentrations were aligned with gas volumes by measuring the time delay for a square-wave bolus of gas passing the turbine to the resulting changes in fractional gas concentrations as measured by the mass spectrometer. Data were collected every 20 ms and transferred to a computer which aligned concentrations with volume information to build a profile of each breath. The measurement for each breath began with the inspiration and concluded with the expiration, thus

enabling to capture the pulmonary gas concentrations at the end of each breath hold.

Legitimacy of using the $P_{ET}O_2$ and $P_{ET}CO_2$ post-breath hold values as a reflection of pulmonary capillary gas pressures

Under sub-lactate threshold exercise intensities, end-tidal partial pressures have been shown to be in equilibrium with that of arterial partial pressures (Suskind et al. 1950). However, during heavy-intensity exercise, ventilatory buffering results in increased breathing frequencies that reduce alveolar–capillary diffusion times for O_2 and CO_2 giving rise to exaggerated higher and lower end-tidal pressures, respectively, than that of pulmonary capillary pressures (Whipp 1977). Since $P_{ET}O_2$ and $P_{ET}CO_2$ during the breath holds of the present study were determined from the last inspiration before the breath hold and the expiration occurring 5 s later, the effects of faster breathing frequencies on the O_2 and CO_2 pulmonary arterial and alveolar end-tidal pressure equilibrium would not evolve. It follows that the lower $P_{ET}O_2$ (hypoxemia) and higher $P_{ET}CO_2$ (hypercapnia) in the breath following the breath holds during CONT-BH and Fartlek-BH would accurately reflect pulmonary arterial pressures.

Muscle deoxygenation [HHb], tissue hemoglobin saturation ($S_{at}O_2$), and total hemoglobin ([Hb_{tot}]) data collection via near-infrared spectroscopy (NIRS)

During all exercise bouts, the vastus lateralis of the quadriceps muscle was monitored continuously using near-infrared spectroscopy (NIRS; Oxiplex TS, model 95,205, ISS, Champaign, IL). The NIRS system was arranged as a single channel consisting of eight laser diodes operating at two wavelengths (690 and 828 nm, 4 at each wavelength) that were pulsed in a rapid succession (frequency modulation of laser intensity was 110 MHz) and a photomultiplier tube. The lightweight plastic NIRS probe (connected to laser diodes and a photomultiplier tube by optical fibers) consisted of two parallel rows of light-emitter fibers and one detector fiber bundle; the source-detector separations for this probe were 2.0, 2.5, 3.0, and 3.5 cm for both wavelengths. The probe was placed on the muscle belly midway between the lateral epicondyle and greater trochanter of the femur; it was secured in place with an elastic strap tightened to prevent movement. The outline of the probe was marked with a permanent marker for future tests. This placement allowed for accurate and continuous measurement of absolute concentration changes (μM) in muscle deoxyhemoglobin ([HHb]), total hemoglobin concentration ([Hb_{tot}]), and tissue hemoglobin saturation ($S_{at}O_2$). The location of measurement was covered with an optically dense, black

vinyl sheet, to minimize the intrusion of extraneous light. The thigh was wrapped with elastic bandages to further minimize the intrusion of extraneous light and movement of the probe. NIRS measurements started 60 s before each test and were collected continuously throughout the entire duration of each test.

The NIRS instrument was calibrated at the beginning of each testing session following a warm-up period of 10 min. The calibration was done with the probe placed on a calibration block (phantom), with absorption (μ_A) and reduced scattering coefficients (μ_S) previously measured; thus, correction factors were determined and were automatically implemented by the manufacturer's software for the calculation of the μ_A and μ_S for each wavelength during the data collection. Calculation of [HHb] reflected continuous measurements of μ_S throughout each testing session (i.e., constant scattering value not assumed). Data were stored online at an output frequency of 25 Hz but were reduced to 1 s bins.

Heart rate was measured continuously by a heart rate monitor (Polar Electro T34) using PowerLab (ML132/ML880, ADInstruments, Colorado Springs, CO) and was calculated (using a 5 s rolling average) based on the RR interval. Data were recorded using LabChart version 6.1 (ADInstruments) on a separate computer.

Arterialized-capillary blood lactate concentrations ($[La^-]$) were measured 3 min before and 3 min after each test. Prior to the use of the lancet, a topical thermogenic (Finalgon, Boehringer Ingelheim) was applied onto the left index finger and then sterilized with a rubbing alcohol swab for each test. Blood was revealed using an ACCU-CHEK Safe-T-Pro Plus sterile, single-use lancing device, and the sample was immediately analyzed by SensLab GmbH Lactate SCOUT arterialized-capillary lactate analyzer ($mmol\ L^{-1}$). Latex gloves were worn by the attending researcher.

Data analysis

Breath-by-breath gas exchange data were filtered by removal of aberrant data points that lay $3 \pm$ above and below the local mean (Lamarra et al. 1987). Data for each protocol were then interpolated linearly to 1 s intervals and time-aligned, such that time zero represented the increase from the 20 W cycling period to the heavy-intensity power output. This second-by-second data were then averaged into 5 s bins for statistical analysis and graphing. [HHb] and $[Hb_{tot}]$ data were zeroed with the baseline. [HHb] and $[Hb_{tot}]$ baseline values determined by the average of 60 s before the step-transition to yield the changes in concentrations respective to their baseline values ($\Delta[HHb]$ and $\Delta[Hb_{tot}]$), and raw $S_{at}O_2$ data were reported. The adjustment of normalized-[HHb]-to- $\dot{V}O_2$ ratio ($\Delta[HHb]/\dot{V}O_2$) was obtained by zeroing and normalizing both the [HHb] and $\dot{V}O_2$ data as percent changes, as previously described (Murias et al. 2011). The baseline values

were considered “0,” and peak values determined from the final 30 s of the exercise were considered “100%”. Analysis of the mean data of the different physiological measures was limited from 120 s to end exercise to eliminate the initial $\dot{V}O_2$ kinetic response. Fluctuations in $\dot{V}O_2$, and $\dot{V}CO_2$, were analyzed by comparing the first 25 s to the last 5 s of each 30 s cycle, whereas $P_{ET}O_2$, $P_{ET}CO_2$, $\Delta[HHb]$, and $\Delta[HHb]/\dot{V}O_2$ were analyzed by comparing the first 20 s to the last 10 s of each 30 s cycle.

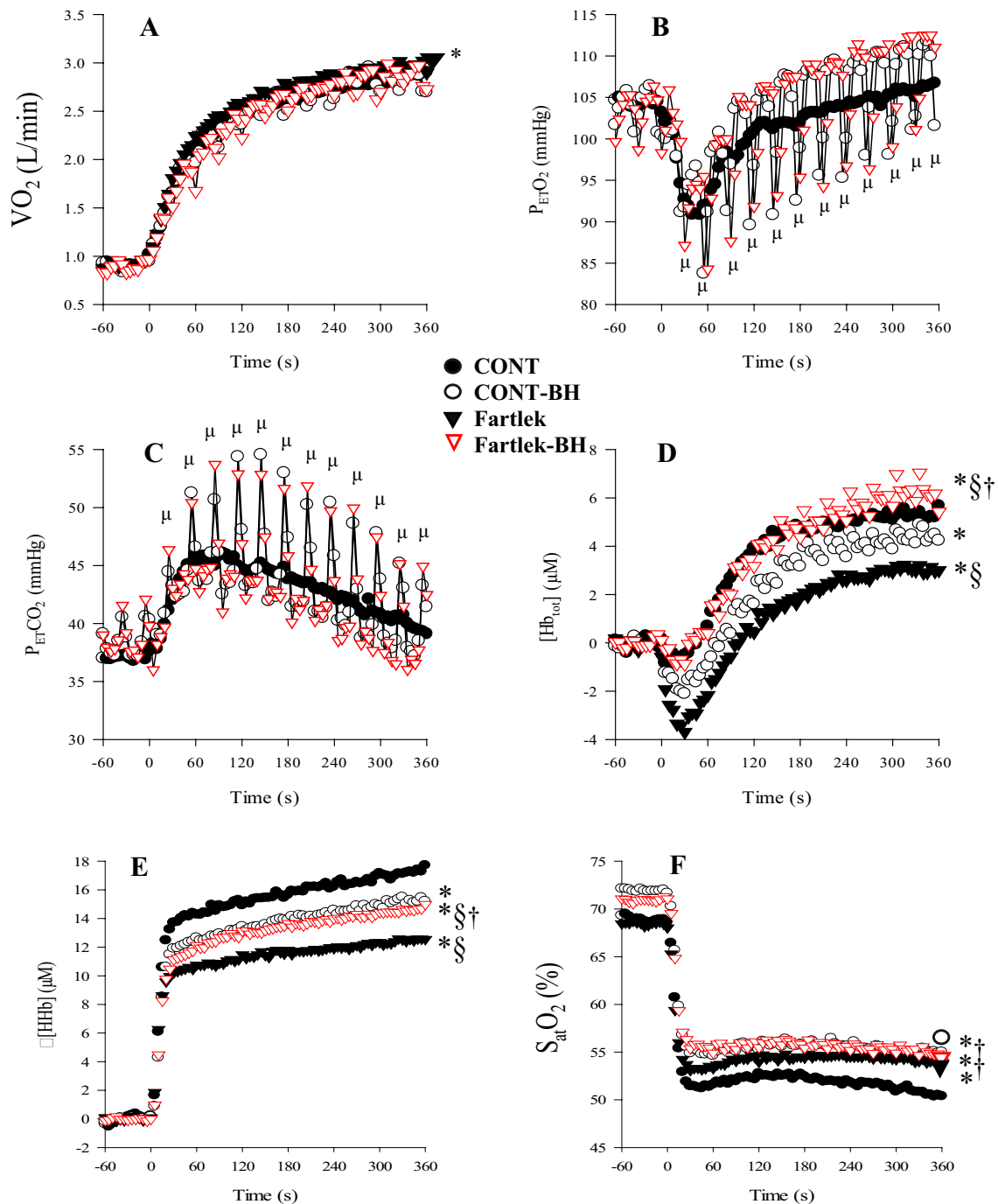
The on-transient of each $\dot{V}O_2$ profile was modeled with the following mono-exponential function:

$$y(t) = y_{BSL} + A_p \times (1 - e^{-(t-TD)/\tau}). \quad (1)$$

In this equation, $y(t)$ is the value of the dependent variable at any time during the transition, y_{BSL} is the pre-transition baseline value, A_p is the steady-state increase in y above the baseline value, τ is the time constant of the response or the time for y to increase to 63% of the new steady-state, and TD is the time delay. The details of the fitting procedure are described elsewhere (Keir et al. 2014). Briefly, the Levenberg–Marquardt algorithm was applied to find the minimum sum of squared residuals between the mono-exponential function and the experimental data using specialized software (Origin 8.5; OriginLab, Northampton, MA). The phase I–phase II transition was determined by examining the change in τ and CI_{95} of the fitting window from the end of exercise, and demarcated as the point at which there was a significant increase in τ and/or CI_{95} closer to the onset of exercise. The end of the phase II fitting window was determined by examining the change in τ , CI_{95} , χ^2 , and plotted residuals in response to progressive increases at the end of the fitting window. The point at which there was a systematic increase in τ , CI_{95} , and χ^2 was considered as the end of phase II.

Statistics

Analysis of the mean results ($n = 10$) from 120 to 360 s of $\dot{V}O_2$, $\dot{V}CO_2$, $\dot{V}_E$, $P_{ET}O_2$, $P_{ET}CO_2$, $[La^-]$, [HHb], $S_{at}O_2$, and $[Hb_{tot}]$ between exercise conditions (CONT, CONT-BH, Fartlek, and Fartlek-BH) were calculated using one-way repeated-measures (RM) ANOVA. $\dot{V}O_2$ kinetics (τ) between conditions was also compared using one-way repeated-measures (RM) ANOVA. To determine if there were oscillations of $\dot{V}O_2$, $\dot{V}CO_2$, $P_{ET}O_2$, $P_{ET}CO_2$, [HHb], and $HHb/\dot{V}O_2$ over the 30 s cycles induced by CONT, CONT-BH, Fartlek, or Fartlek-BH, all 30 s cycles of each participant for each condition were divided into 25 s and 5 s bins between 120 and 360 s. A mean of these 25 s and 5 s periods was then calculated for each participant for each condition. A two-way repeated-measures ANOVA was then used to compare physiological responses across time (25 s and 5 s) within the four conditions. Data are



reported as mean $\pm$ SD. Statistical significance was declared when $p < 0.05$.

Results

Summary of the anthropometric characteristics and performance variables assessed during the ramp incremental tests is shown in Table 1.

Mean $\dot{V}O_2$ ($L \cdot min^{-1}$) carbon dioxide production ($\dot{V}CO_2$ $L \cdot min^{-1}$) and minute ventilation ($\dot{V}_E$ $L \cdot min^{-1}$)

Mean $\dot{V}O_2$ ($L \cdot min^{-1}$) from 120 s to the end of exercise was greater in Fartlek than all the other conditions ($p < 0.05$; Fig. 1a). $\dot{V}O_2$ ($L \cdot min^{-1}$) was lower during the 5 s breath hold compared to the 25 s free-breathing period during CONT-BH and Fartlek-BH (Fig. 1a), whereas there

Fig. 1 CONT: filled circles: continuous (continuous exercise at $\Delta 50$: 50% of the difference between the LT and $\dot{V}O_{2peak}$); open circles: CONT-BH: open circles: (continuous exercise at $\Delta 50$ with repeated 25 s free breathing and 5 s breath holds); Fartlek: filled triangles: (repeated 25 s $\Delta 50$ and 5 s free-breathing sprints—peak work rate attained during ramp incremental test); Fartlek-BH: red open triangles: (repeated 25 s free breathing $\Delta 50$ and 5 s sprints with breath holds). **a** Mean pulmonary oxygen uptake ($\dot{V}O_{2p}$) from 120 s to the end of exercise was greater in Fartlek than all the other conditions ($p < 0.05$). No other mean differences were observed ($p > 0.05$). One-way repeated-measures ANOVA (one-way RM ANOVA). Differences in $\dot{V}O_{2p}$ between the 25 s free breathing (FB) and the 5 s BH from 0 to 360 s ($p < 0.05$) two-way RM ANOVA. **b** Mean end-tidal partial pressures of oxygen ($P_{ET}O_2$). μ Differences in $P_{ET}O_2$ between the 20 s FB and the 5 s BH and the following 5 s of FB from 0 to 360 s ($p < 0.05$). Two-way RM ANOVA. **c** Mean end-tidal partial pressures of carbon dioxide $P_{ET}CO_2$; μ Differences in $P_{ET}CO_2$ between the 20 s FB and the 5 s BH and the following 5 s of FB from 0 to 360 s ($p < 0.05$) two-way RM ANOVA. **d** Mean total hemoglobin ($\Delta[Hb_{tot}]$) from 120 s to the end of exercise during CONT-BH and Fartlek was lower than CONT, and Fartlek was lower than Fartlek-BH ($p < 0.05$). **e** Change in deoxygenated hemoglobin concentration ($\Delta[HHb]$) from baseline values from 120 s to the end of exercise was different between all the conditions ($p < 0.05$). **f** Mean tissue hemoglobin saturation ($S_{at}O_2$) from 120 s to the end of exercise during CONT-BH and Fartlek-BH were greater than CONT ($p < 0.05$). *Different from CONT, † different from CONT-BH, and ‡ different from Fartlek

were no fluctuations in CONT and Fartlek over this same period (Fig. 1a).

The mean $\dot{V}CO_2$ ($L \min^{-1}$) from 120 s to the end of exercise was greater in Fartlek (3.43 ± 0.16) than CONT (3.12 ± 0.38) and lower in Fartlek-BH (3.28 ± 0.35) than Fartlek ($p < 0.05$) but similar in CONT-BH (3.16 ± 0.38) to CONT. Mean $\dot{V}CO_2$ fluctuations were observed during CONT-BH and Fartlek-BH (Fig. 1c, $p < 0.05$) but not during Fartlek (Fig. 1c, $p > 0.05$).

The mean $\dot{V}_E$ ($L \min^{-1}$) from 120 s to the end of exercise was greater in CONT-BH, Fartlek, and Fartlek-BH than CONT, greater in Fartlek-BH than CONT-BH, and lower in Fartlek-BH than Fartlek ($p > 0.05$; Table 2).

$\dot{V}O_2$ kinetics

$\tau\dot{V}O_2$ was greater in Fartlek-BH than CONT and Fartlek (Table 2; $p < 0.05$), whereas CONT-BH, Fartlek, and CONT were similar (Table 2; $p > 0.05$) and Fartlek was different than CONT-BH and Fartlek-BH (Table 2; $p < 0.05$).

$P_{ET}O_2$ mmHg and $P_{ET}CO_2$

$P_{ET}O_2$ was lower ($p < 0.05$) over the 10 s periods that began with the inspiration of the BH, compared to resting as well as the following 20 s free-breathing periods (Fig. 1b). Similar temporal but inverted mean $P_{ET}CO_2$ fluctuations were observed during CONT-BH and Fartlek-BH (Fig. 1c). No

fluctuations were observed during CONT or Fartlek ($p > 0.05$). Fartlek data on Fig. 1b, c have been omitted for clarity.

Change in mean total hemoglobin concentration ($\Delta[Hb_{tot}]$), change in mean deoxygenated hemoglobin concentration ($\Delta[HHb]$), mean tissue hemoglobin saturation ($S_{at}O_2$), and $[HHb]$ -to- $\dot{V}O_2$ ratio

The mean $\Delta[Hb_{tot}]$ from 120 s to the end of exercise was higher in Fartlek-BH than all the other conditions and Fartlek was the lowest ($p < 0.05$; Fig. 1d). CONT-BH was lower than CONT ($p < 0.05$; Fig. 1d).

The mean $\Delta[HHb]$ changes from baseline values from 120 s to the end of exercise were the highest in CONT compared to the other conditions ($p < 0.05$; Fig. 1e). However, both breath hold conditions were higher than Fartlek ($p < 0.05$; Fig. 1e).

The $S_{at}O_2$ from 120 s to the end of exercise was the lowest during CONT compared to all conditions ($p < 0.05$; Fig. 1f). However, both breath hold conditions were higher than Fartlek ($p < 0.05$; Fig. 1f).

Mean $\Delta[HHb]/\dot{V}O_2$ from 0 to 360 s was similar between all conditions ($p > 0.05$; Fig. 2). However, fluctuations in mean $\Delta[HHb]/\dot{V}O_2$ during CONT-BH and Fartlek-BH were observed ($p < 0.05$; Fig. 2).

Arterialized-capillary lactate concentration ($[La^-]$)

The mean pre-exercise $[La^-]$ across all conditions was similar (Table 2; $p > 0.05$) and mean post-exercise $[La^-]$ in Fartlek-BH was greater than all other conditions (Table 2; $p < 0.05$).

For your information, numerical data from the statistical analysis on the figures for select variables have been included in tabular form as Appendix 1.

Discussion

The novel purpose of this study was to compare and contrast select cardiorespiratory and metabolic responses during heavy-intensity constant-load cycle ergometer exercise (CONT), to three differing intermittent heavy-intensity exercise protocols. These protocols included: (1) repeated 5 s periods of breath holding every 30 s (CONT-BH), (2) repeated 5 s periods of higher power output (sprints) every 30 s (Fartlek), and (3) a combination of the sprints and the breath holds (Fartlek-BH).

The main findings demonstrated that, compared to CONT, (1) CONT-BH resulted in similar mean $\dot{V}O_2$, $\dot{V}O_2$ kinetics, and $[La^-]$, (2) Fartlek resulted in increased mean $\dot{V}O_2$ yet unchanged $\dot{V}O_2$ kinetics and $[La^-]$, whereas

Table 2 Summary of selected physiological parameters collected during CONT, CONT-BH, Fartlek, and Fartlek-BH

	CONT	CONT-BH	Fartlek	Fartlek-BH		CONT	CONT-BH	Fartlek	Fartlek-BH
Mean PO (W)	218	218	234 ^a	234 ^a	$\dot{V}_E$ (L min ⁻¹)	90	98 ^a	105 ^{a,b}	101 ^{a, c}
SD	30	30	33	33	SD	7	16	10	19
$\dot{V}O_{2p}$ BSL (L min ⁻¹)	0.92	0.90	0.93	0.90	Pre [La ⁻] (mM)	2.0	1.8	1.7	1.7
SD	0.04	0.03	0.04	0.05	SD	0.5	0.3	0.4	0.4
$\tau\dot{V}O_{2p}$ (s)	49	53	44 ^{b, d}	62 ^{a, c}	Post [La ⁻] (mM)	9.0 ^d	10.0 ^d	9.9 ^d	11.5
SD	12	11	8	19	SD	2.3	2.4	2.0	2.7

Data analyzed by one-way repeated-measures ANOVA. Values are given as means $\pm$ SD. Mean data are from 120 to 360 s

CONT continuous exercise at $\Delta 50$ with free-breathing, *CONT-BH* continuous exercise at $\Delta 50$ (power output at 50% of the difference between the LT and $\dot{V}O_{2peak}$) with repeated 30 s cycles comprised of 25 s free-breathing periods and 5 s breath holds, *Fartlek* continuous exercise with repeated 30 s cycles comprised of 25 s $\Delta 50$ and 5 s sprints (peak power output attained during ramp incremental test) with free breathing, *Fartlek-BH* repeated 30 s cycles comprised of 25 s $\Delta 50$ with free-breathing and 5 s sprints combined with a breath hold, *Mean PO* mean power output, $\dot{V}O_{2p}$ BSL baseline during 4 min of 20 W cycling, [La⁻] arterialized-capillary lactate concentration, $\dot{V}_E$ mean ventilation

^aDifferent from CONT

^bDifferent from CONT-BH

^cDifferent from Fartlek

^dDifferent from Fartlek-BH

(3) Fartlek-BH resulted in unchanged mean $\dot{V}O_2$, slower $\dot{V}O_2$ kinetics, and increased [La⁻].

CONT-BH

Despite the fluctuations in $P_{ET}O_2$ and $P_{ET}CO_2$, consequent to the breath holds (Fig. 1b, c), mean $\dot{V}O_2$ (Fig. 1a) and $\dot{V}CO_2$ were unchanged. Mechanistically, the transient increase in $P_{ET}CO_2$ (Fig. 1c) would have evoked the observed hyperventilatory response (Table 2) via peripheral and central chemoreceptor input to the respiratory center located in the medulla (Whipp and Davis 1979). Furthermore, the observed decrease in $\dot{V}O_2$ over the duration of the breath hold (Fig. 1a and Table 3), concomitant with the unchanging energy demand, would require an increased contribution from anaerobic phosphorylation to maintain the work rate. This increase in anaerobic glycolytic contribution has been shown to increase H^+ concentration (Belfry et al. 2012b) and is linked to a concurrent increase in [La⁻] (Kowalchuk et al. 1988). As H^+ is substrate for the carbonic anhydrase reaction and CO_2 is the product, it follows that the observed increases of both $\dot{V}CO_2$ and the consequential increase of $\dot{V}_E$ (ventilatory buffering) during CONT-BH of the present study, has reduced $[H^+]$ and the associated [La⁻] accumulation (Table 2) (Kowalchuk et al. 1988) whilst sustaining $\dot{V}O_2$.

Compared to the CONT-BH condition of the present study, others (Kume et al. 2013; Woorens et al.) utilizing longer duration hypoxia inducing protocols (15 s of breath holding per min and 45 s of hypoventilation per min, respectively), observed either no change in $\dot{V}CO_2$ and $\dot{V}_E$, or, in the latter study, gas exchange was not measured. Presumably, unlike CONT-BH of the current study, the longer duration breath holds utilized in the Woorens et al.'s study;

restricted ventilatory buffering to such an extent that an increase of $\dot{V}CO_2$ elimination was not possible. There were no observations of blood lactate in that study. The increased blood lactates observed in the hypoventilation condition of the Kume et al.'s study suggest a breath hold-induced reduction of ventilatory buffering that was not observed during the CONT-BH condition of the present study.

Lindholm and Linnarsson (2002) and others (Ferretti et al. 1991; Andersson and Schagatay 1998) have suggested that breath holding results in O_2 conservation at the level of the muscle that preserves O_2 availability for the brain and the central nervous system. Others (Hong et al. 1971) have not. The decreased $\Delta[Hb_{tot}]$ (Fig. 1d) and deoxygenated hemo/myoglobin ($\Delta[HHb]$) (Fig. 1e), and increased O_2 saturation of the hemo/myoglobin ($S_{at}O_2$) (Fig. 1f), coupled with the reduction of $\dot{V}O_2$ during the breath holds (Fig. 1a) observed in the current study, suggest a similar O_2 conservation mechanism at work at the level of the muscle. These changes suggest the presence of the previously observed breath hold-induced reduction in cardiac output (Andersson et al. 2004).

This apneic response is dissimilar to that observed during hypoxia under free-breathing conditions wherein increased muscle deoxygenation increases O_2 availability at the muscle (Subudhi et al. 2007). Despite this conservation of O_2 during the CONT-BH condition, the increased $\dot{V}_E$ (Table 2) during the 25 s free-breathing periods has underpinned the sustained $\dot{V}O_2$ from CONT to CONT-BH.

The novel observations of the transient breath hold-induced reductions in $\dot{V}O_2$, (Table 3) but not $\Delta[HHb]$ in the present study (Fig. 1e), have buttressed the recurring zeniths of $\Delta[HHb]/\dot{V}O_2$ (Fig. 2). Similar temporal changes in $\dot{V}O_2$ have been also observed during identical durations, but as

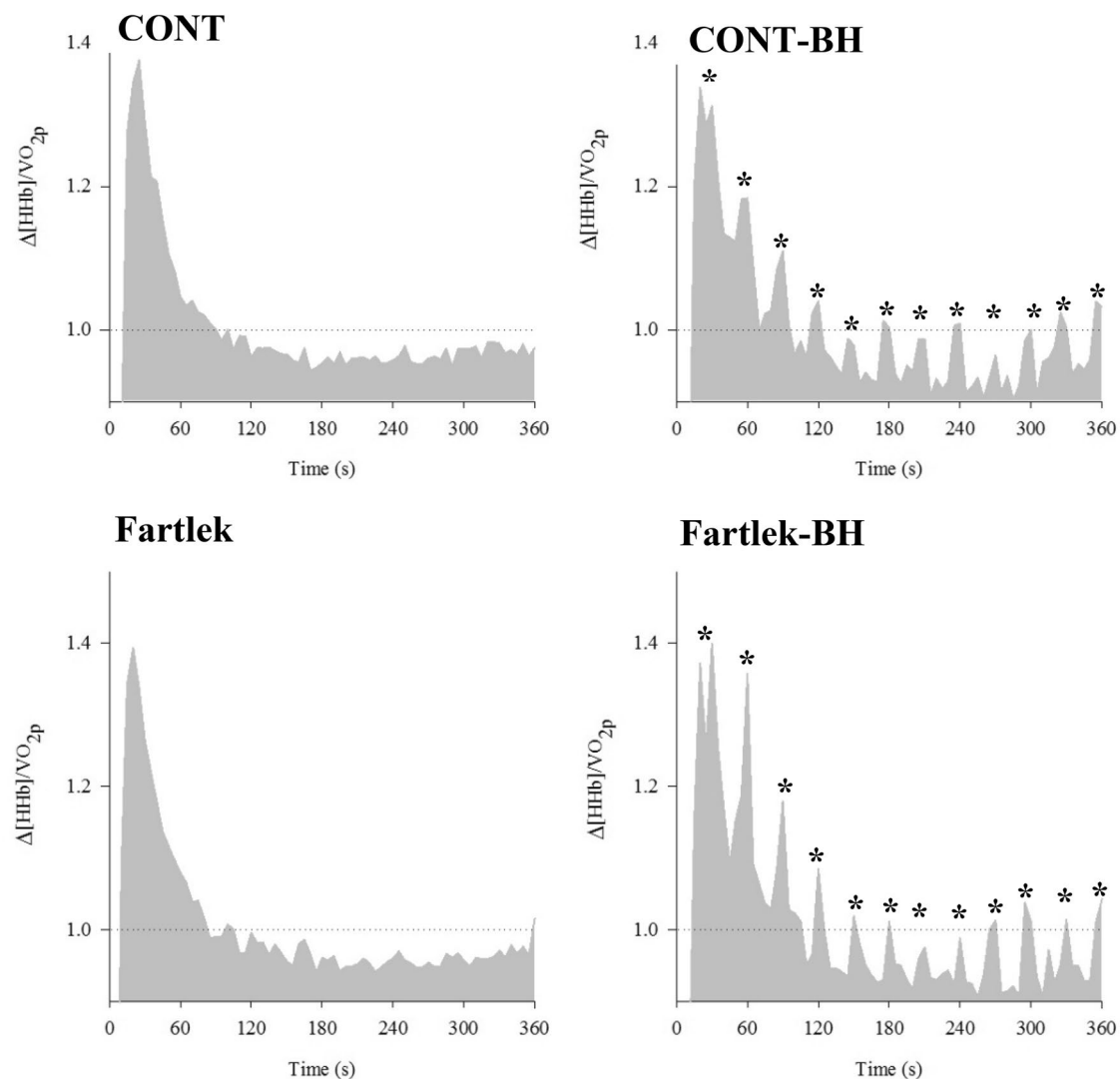


Fig. 2 Mean adjustment of normalized [HHb]-to- $\dot{V}O_{2p}$ ratio ($\Delta[HHb]/\dot{V}O_{2p}$) during CONT (continuous exercise at $\Delta 50$: 50% of the difference between the lactate threshold and $\dot{V}O_{2peak}$), CONT-BH (continuous exercise at $\Delta 50$ with repeated 25 s free-breathing and 5 s breath holds), Fartlek (repeated 25 s $\Delta 50$ and 5 s free-breathing

sprints—peak power output attained during ramp incremental test), and Fartlek-BH (repeated 25 s free-breathing $\Delta 50$ and 5 s sprints with breath holds). * $\Delta[HHb]/\dot{V}O_{2p}$ during the last 5 s were greater than the nadir values during the 25 s periods over the 30 s cycles ($p < 0.05$) two-way RM ANOVA

a consequence of reduced O_2 demand (Belfry et al. 2012a). This demonstrates that abrupt $\dot{V}O_2$ kinetic adjustments to both a reduction in O_2 demand (Belfry et al. 2012a), as well as decreased O_2 availability during heavy-intensity exercise, as observed in the present study, are possible.

The previously observed slowing of $\dot{V}O_2$ kinetics during exercise in hypoxia (Springer et al. 1991; Hughson and Kowalchuk 1995; Engelen et al. 1996; DeLorey et al. 2004) and longer duration breath holds (Hoffmann et al. 2005; Schagatay et al. 2007; Kume et al. 2013) was not hypothesised in the CONT-BH condition of the current study, and, subsequently, not observed (Table 2). It is suggested that the combined effects of the brevity of the breath holds and

the increased $\dot{V}_E$ during the 25 s free-breathing periods of CONT-BH (Table 2) have compensated for the reduced breathing opportunities during CONT-BH. Thus, O_2 delivery and associated $\dot{V}O_2$ kinetics have been preserved.

From a backstroke swimming perspective, these data suggest that the oxidative phosphorylation demand required to maintain the intensity of the kick throughout the underwater phase, and on into the free-breathing surface swimming phases, can be supported via increased $\dot{V}_E$. Whether the additional cardiorespiratory demands of full stroke swimming upon surfacing (arms and legs) could be met by a further increase in $\dot{V}_E$ is still in question. However, the maximal $\dot{V}_E$ observed at $\dot{V}O_{2peak}$ during the incremental test

($24 \text{ L} \times 5 \text{ s}^{-1}$), compared to the $\dot{V}_E$ during each 5 s period of free breathing during CONT-BH ($19 \text{ L} \times 5 \text{ s}^{-1}$), suggests some ventilatory accommodation would be possible.

Fartlek

Our previous work on intermittent exercise patterns has shown that regular insertions of short-recovery periods (5 s and 3 s) during heavy-intensity cycling exercise resulted in decreased mean $\dot{V}O_2$, $\dot{V}CO_2$ and $\Delta[\text{HHb}]$ (Belfry et al. 2012a; McCrudden et al. 2017). Conversely, in the current study, we hypothesised that if these short-recovery periods were replaced with 5 s sprints, an increase in mean $\dot{V}_E$, $\dot{V}O_2$, $\dot{V}CO_2$, $[\text{La}^-]$, and $\Delta[\text{HHb}]$ compared to CONT would result. Indeed, the insertions of 5 s sprints resulted in increased mean $\dot{V}O_2$ (Fig. 1a), $\dot{V}CO_2$ (“Results”), and $\dot{V}_E$ (Table 2). However, unexpectedly, no change in $[\text{La}^-]$ was observed (Table 2). It would still be expected that the repeated transitions from 218 W of the 25 s period, to the 314 W during the 5 s sprint, would have required both PCr-derived phosphorylation and glycolytic phosphorylation contributions (Belfry et al. 2012b) from the increased type II fiber recruitment (Gollnick et al. 1974; Nuutinen et al. 1982). However, it is suggested that this increased anaerobic glycolytic contribution and concomitant increased $[\text{La}^-]$ has been eliminated by the increased $\dot{V}CO_2$ (“Results”) and associated $\dot{V}_E$ (Table 2).

If the standard O_2 cost of work ($\sim 10 \text{ mL } O_2 \text{ min}^{-1} \text{ W}^{-1}$) (Poole et al. 1992) is used, our data suggest that the increased mean O_2 demand of Fartlek ($160 \text{ mL } O_2$) has essentially been met by the observed increase in oxidative phosphorylation as reflected by the increased $\dot{V}O_2$ ($140 \text{ mL } O_2$) (Fig. 1a).

The decreased $\Delta[\text{HHb}]$, as well as the $\Delta[\text{Hb}_{\text{tot}}]$, and increased $S_{\text{at}}O_2$, coupled with the increased $\dot{V}O_2$ (Fig. 1a) suggests a dramatic enhancement in muscle blood flow distribution. This favourable microvascular response has also been observed during similar higher power output contractions (Walløe and Wesche 1988; Saltin et al. 1998).

Finally, $\dot{V}O_2$ kinetics were similar between Fartlek and CONT (Table 2). It is suggested that the expected increase in anaerobic glycolytically derived H^+ , by which $\dot{V}O_2$ kinetics would have been slowed (Linnarsson et al. 1974; Hughson and Kowalchuk 1995), was offset by the observed increase in ventilatory buffering, as well as the improved muscle blood flow distribution and O_2 delivery (Peronnet and Aguilaniu 2006). It is suggested that the faster $\dot{V}O_2$ kinetics of Fartlek relative to CONT-BH evolves from an amalgamation of (1) the transient, sprint related, decreases in PCr and increased ADP (Belfry et al. 2012b) which would speed $\dot{V}O_2$ kinetics, and (2) the absence of the slowing effects that have been linked to increased $[\text{La}^+]$ (Table 2) and the transient hypoxaemia reflected by the decreased $P_{\text{ET}}O_2$ of CONT-BH (Fig. 1b) (Rossiter et al. 2002).

From a backstroke swimming perspective, these Fartlek data suggest that if, by design, the underwater kicking phase was kept as short as possible, by surfacing immediately after a turn, then the aforementioned reserve of $\dot{V}_E$ could be utilized to sustain the increased O_2 demand associated with either an increase in kicking intensity and/or the additional O_2 requirements of the arm work during full stroke swimming.

Fartlek breath hold (Fartlek-BH)

The reduced duration of breathing opportunities during Fartlek-BH, compared to Fartlek, resulted in decreased mean $\dot{V}_E$ (Table 2). It is suggested that this reduction has impacted the efficacy of ventilation on buffering and $\dot{V}O_2$ during Fartlek-BH, as $[\text{La}^-]$ increased (Table 2) and $\dot{V}O_2$ decreased (Fig. 1a). This is consistent with the standard calculations of oxygen uptake, whereby the observed decrease in ventilation from Fartlek to Fartlek-BH would decrease oxygen uptake. This moderation of the oxidative phosphorylation contribution during Fartlek-BH has been replaced by an increased contribution from anaerobic glycolysis as reflected by the increased $[\text{La}^-]$ (Peronnet and Aguilaniu 2006). Similar to CONT-BH, Fartlek-BH (Table 3) shows temporal oscillations in $\dot{V}O_2$ during the breath hold. This phenomena has been observed during 5 s recovery periods of reduced O_2 demand during intermittent exercise (Belfry et al. 2012a). Taken together this suggests that if either a 5 s period of apnea induced O_2 conservation or O_2 demand is present, oxidative phosphorylation will decrease accordingly.

Moreover, the increased $\Delta[\text{Hb}_{\text{tot}}]$ (Fig. 1d) and $S_{\text{at}}O_2$ (Fig. 1f), coupled with the reduced $\Delta[\text{HHb}]$ (Fig. 1e) compared to CONT, intimates a similar breath hold provoked O_2 conservation response as was observed during CONT-BH. Notwithstanding, the increased microvascular blood flow and O_2 availability, a concurrent and proportional increase in $\dot{V}O_2$ did not occur. This increase in $\Delta[\text{Hb}_{\text{tot}}]$ reflecting increased blood flow has been associated with the observed increase in $[\text{La}^-]$ that has linked to the activation of cyclic guanine monophosphate (cGMP) within smooth muscle cells of the muscle microvasculature (Chen et al. 1996). However, this increase in blood flow during exercise under hypercapnic conditions (DeLorey et al. 2004) has been shown to reduce hemoglobin capillary and diffusion times, and a reduction of reduced O_2 capillary to muscle diffusion has been observed. This O_2 conservation mechanism in the face of the increased O_2 demand would account for the unchanged $\dot{V}O_2$ during Fartlek-BH.

Finally, the slower $\dot{V}O_2$ kinetics during Fartlek-BH compared to both CONT and Fartlek (Table 2) is consistent with the previous observations of slowed $\dot{V}O_2$ kinetics during exercise in hypoxia (Springer et al. 1991; Hughson and Kowalchuk 1995; Engelen et al. 1996; DeLorey et al. 2004). Since there was no evidence of hypoxia in the present

study, it is suggested that the observed increase in $[La^-]$ resulting from increased Type II muscle fiber recruitment (Gollnick et al. 1974; Murphy et al. 1989; Nuutinen et al. 1982) is responsible for the slowed kinetics. Furthermore, the addition of the breath hold-induced O_2 conservation phenomena, occurring at the level of the muscle, would also have contributed to the slowed $\dot{V}O_2$ kinetics.

In summary, and as hypothesised, the combination of the breath hold and sprint conditions resulted in increased arterialized-capillary lactate, slowed $\dot{V}O_2$ kinetics, and unchanged $\dot{V}O_2$ compared to both the CONT-BH and Fartlek conditions.

Finally, even without the additional arm work associated with the full stroke swimming model, the cardiorespiratory response to the Fartlek-BH perturbation was not sufficient to facilitate the required increase in ventilation to facilitate buffering and/or support the required $\dot{V}O_2$. Unless a swimmer's kicking efficiency (metabolic cost to kicking velocity) was much superior to their competitors and a significant advantage was to be expected using this strategy, the underwater sprint kicking tactic would not be recommended.

Summary of practical applications

Competitive backstroke swimmers may utilize different breath hold and power output strategies when performing the underwater kicking phase after turns. Moreover, some swimmers may achieve a faster velocity at similar power outputs due to greater kicking efficiencies. For these athletes, despite the increased metabolic stress and quicker time to fatigue of combining sprint kicking to the breath holds this perturbation may be advantageous. Conversely, it is suggested that those less efficient kickers be advised to limit their underwater kicking durations as there is no physiological or swimming velocity advantage to be gained from this

strategy. Experimentation to determine what underwater kicking strategy works best for each swimmer, coupled with a knowledge of their competition's underwater capacities would be vital for achieving their optimal placings.

It is important to recognise that the heretofore discussion, of the “on land” cardiorespiratory responses to the different perturbations of this study, does not account for the differential cardiorespiratory responses associated with the hydrostatic pressure changes experienced during underwater kicking at a depth of ~1 m. These include an increase in venous return, which translates into greater cardiac outputs at all intensities of exercise (Christie et al. 1990), as well as an “engorgement” of the pulmonary vasculature (Muth et al. 2005). The reduced muscle $[HHb]$, $[Hbtot]$, and increased $S_{at}O_2$ observed in the breath hold conditions of the present study, implicating O_2 conservation, would limit or negate, the increased O_2 delivery associated with the increased hydrostatic immersion pressure at the level of the muscle. On the other hand, it is also suggested that the “engorgement” of the pulmonary vasculature during the breath hold would increase gas pressures in the pulmonary capillaries. This would result in decreased O_2 diffusion from the lung to the pulmonary capillaries and reduce $S_{at}O_2$. In theory, this would mitigate any O_2 conservation effects of the breath hold. Regardless, further study to elucidate these specific effects under immersion and full stroke swimming conditions is required.

Conclusion

To our knowledge, this is the first study to examine the physiological responses to the insertions of intermittent 5 s breath holds and/or sprints during heavy-intensity cycle ergometer exercise. We conclude that the independent stimuli of the breath holds and the sprints during high-intensity exercise are accommodated successfully. However, the additional metabolic stress of combining these perturbations results in a disruption of homeostasis.

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Author contributions GRB and DL conceived and designed the study; GRB, DL, and JK collected and analyzed the data; GRB, DL, and JK interpreted the results and drafted the manuscript; GB and GDM edited the manuscript.

Table 3 Summary of fluctuations in $\dot{V}O_{2p}$ and $\dot{V}CO_{2p}$ between, 25 s and 5 s, intervals within the 30 s cycles CONT-BH, Fartlek, and Fartlek-BH

	CONT-BH		Fartlek-BH	
	25 s	5 s	25 s	5 s
$\dot{V}O_{2p}$ (L min ⁻¹)	2.67	2.25 ^a	2.70	2.20 ^a
SD	0.40	0.50	0.50	0.50
$\dot{V}CO_{2p}$ (L min ⁻¹)	3.13	2.03 ^a	3.29	2.07 ^a
SD	0.80	0.54	0.90	0.70

Data analyzed by two-way repeated-measures ANOVA. Data from 0 to 360 s. Values are given as means $\pm$ SD

25 s the first 25 s of each 30 s cycle, 5 s the last 5 s of each 30 s cycle, CONT-BH continuous breath hold, Fartlek-BH fartlek breath hold, $\dot{V}O_{2p}$ mean pulmonary oxygen uptake, $\dot{V}CO_{2p}$ mean carbon dioxide production

^a25 s different from 5 s

Appendix 1

See Table 4.

Table 4 Summary of physiological parameters collected during CONT, CONT-BH, Fartlek, and Fartlek-BH

	CONT	CONT-BH	Fartlek	Fartlek-BH		CONT	CONT-BH	Fartlek	Fartlek-BH
$\dot{V}O_{2p}$ (L min ⁻¹)	2.71	2.73	2.85 ^a	2.73 ^c	Δ [HHb] (uM)	16.3	14.4 ^a	12.0 ^{a,b}	13.9 ^{a,b,c}
SD	0.12	0.14	0.12	0.14	SD	0.7	0.6	0.4	0.5
$P_{ET}O_2$ (mmHg)	104	105	108 ^a	106 ^{a,b}	$S_{at}O_2$ (%)	51.7	55.5 ^{a,c}	54.4 ^a	55.5 ^{a,c}
SD	2	5	2	5	SD	0.7	0.5	0.2	0.5
$P_{ET}CO_2$ (mmHg)	42	43	40 ^a	42 ^{b,c}	Δ [HHb]/ $\dot{V}O_{2p}$	0.99	0.99	1.00	1.00
SD	2	4	3	4	SD	0.13	0.15	0.12	0.15
Δ [Hb _{tot}] (uM)	4.9	3.9 ^a	2.4 ^{a,b}	5.4 ^{a,b,c}					
SD	0.4	0.7	0.7	0.8					

Data analyzed by one-way repeated-measures ANOVA. Values are given as means $\pm$ SD. Mean data are from 120 to 360 s

CONT continuous exercise at $\Delta 50$ with free-breathing, CONT-BH continuous exercise at $\Delta 50$ with repeated 30 s cycles comprised of 25 s free-breathing periods and 5 s breath holds, Fartlek continuous exercise with repeated 30 s cycles comprised of 25 s $\Delta 50$ and 5 s sprints with free-breathing, Fartlek-BH repeated 30 s cycles comprised of 25 s $\Delta 50$ with free-breathing and 5 s sprints combined with a breath hold, $P_{ET}O_2$ mean end-tidal partial pressures of oxygen, $P_{ET}CO_2$ mean end-tidal partial pressures of carbon dioxide, Δ [Hb_{tot}] mean change in total hemoglobin concentration from baseline values, Δ [HHb] mean change in deoxygenated hemoglobin concentration from baseline values, $S_{at}O_2$ mean tissue hemoglobin saturation, Δ [HHb]/ $\dot{V}O_{2p}$ adjustment of normalized [HHb]-to-normalized $\dot{V}O_{2p}$ ratio from 0 to 360 s

^aDifferent from CONT

^bDifferent from CONT-BH

^cDifferent from Fartlek

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